

Practical-7

Aim:

To optimize the performance of an Apache Spark application by applying Caching and Partitioning techniques on a movie ratings dataset and comparing the execution time before and after optimization.

Problem Definition:

You are a Big Data Engineer at MovieFlix Technologies, a movie recommendation company serving millions of users worldwide. The recommendation engine repeatedly processes movie ratings data to generate personalized suggestions. Due to the increasing volume of user interactions, query execution time has become a major concern. Management wants to optimize the Spark application using Caching and Partitioning techniques to reduce execution time and improve scalability.

Objectives:

1. To create a Spark Session and load the movie ratings dataset into a Spark DataFrame.
2. To display and understand the schema of the dataset.
3. To check the initial number of partitions.
4. To calculate the average movie ratings.
5. To measure execution time before optimization.
6. To apply `cache()` and trigger caching using `count()`.
7. To measure execution time after caching.
8. To repartition the dataset and observe the change in the number of partitions.
9. To compare the performance before and after caching and partitioning.
10. To understand the effect of `cache()`, `persist()`, `repartition()`, and `coalesce()` on Spark performance.

Objective/Output:

Tasks: Create Spark Session. Load ratings dataset into Spark DataFrame.
Display dataset schema.

```
This message is shown once a day. To disable it please create the
/home/gitanshu/.hushlogin file.
gitanshu@Parth:~$ mkdir -p spark_practical7
gitanshu@Parth:~$ cd spark_practical7
gitanshu@Parth:~/spark_practical7$ pwd
/home/gitanshu/spark_practical7
gitanshu@Parth:~/spark_practical7$ nano ratings.csv
gitanshu@Parth:~/spark_practical7$ nano practical7.py
gitanshu@Parth:~/spark_practical7$ ls
practical7.py  ratings.csv
gitanshu@Parth:~/spark_practical7$ cd ~/spark_practical7
```

```
gitanshu@Parth:~/spark_practical7$ cat ratings.csv
userId,movieId,rating,timestamp
1,101,4.0,964982703
1,102,3.5,964981247
1,103,5.0,964982224
2,101,4.5,964982931
2,104,3.0,964983815
2,105,4.0,964982400
3,102,4.0,964982176
3,103,4.5,964982344
3,106,2.5,964982224
4,101,5.0,964983611
4,104,3.5,964982400
4,107,4.0,964982931
5,102,3.0,964982176
5,105,4.5,964982344
5,108,5.0,964982224
6,103,4.0,964983611
6,106,3.5,964982400
6,109,4.5,964982931
7,101,3.5,964982176
7,107,4.0,964982344
7,110,5.0,964982224
8,104,4.5,964983611
8,108,3.0,964982400
8,109,4.0,964982931
9,105,4.0,964982176
9,106,4.5,964982344
9,110,3.5,964982224
10,101,5.0,964983611
```

Check the current number of partitions.

```

+-----+-----+-----+-----+
|userId|movieId|rating|timestamp|
+-----+-----+-----+-----+
|      1|      101|    4.0|964982703|
|      1|      102|    3.5|964981247|
|      1|      103|    5.0|964982224|
|      2|      101|    4.5|964982931|
|      2|      104|    3.0|964983815|
|      2|      105|    4.0|964982400|
|      3|      102|    4.0|964982176|
|      3|      103|    4.5|964982344|
|      3|      106|    2.5|964982224|
|      4|      101|    5.0|964983611|
+-----+-----+-----+-----+
only showing top 10 rows

Dataset Schema:
root
|-- userId: integer (nullable = true)
|-- movieId: integer (nullable = true)
|-- rating: double (nullable = true)
|-- timestamp: integer (nullable = true)

```

Calculate average movie ratings.

Measure execution time before optimization.

```

+-----+-----+
|movieId|average_rating|
+-----+-----+
|      108|          4.0|
|      101|          4.4|
|      103|          4.5|
|      107|          4.0|
|      102|          3.5|
|      109| 4.333333333333333|
|      105| 4.166666666666667|
|      110|          4.25|
|      106|          3.5|
|      104| 3.666666666666665|
+-----+-----+

```

Execution Time BEFORE Optimization: 1.243778 seconds

Apply cache() on the DataFrame. Trigger caching using count().

```
26/09/28 17:02:41 INFO DAGScheduler: ResultStage 9 ($anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768) finished in 43 ms
26/09/28 17:02:41 INFO DAGScheduler: Job 7 is finished. Cancelling potential speculative or zombie tasks for this job
26/09/28 17:02:41 INFO TaskSchedulerImpl: Canceling stage 9
26/09/28 17:02:41 INFO TaskSchedulerImpl: Killing all running tasks in stage 9: Stage finished
26/09/28 17:02:41 INFO DAGScheduler: Job 7 finished: $anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768, took 48.945238 ms
26/09/28 17:02:41 INFO DAGScheduler: Asked to cleanup jobs for query execution 4
Cache Triggered Successfully!
Total Records: 30
Cache Time: 0.454688 seconds
```

Recalculate average ratings.

```
26/09/28 17:02:41 INFO DAGScheduler: Asked to cleanup jobs for query execution 4
+-----+
|movieId|    average_rating|
+-----+
|    108|             4.0|
|    101|             4.4|
|    103|             4.5|
|    107|             4.0|
|    102|             3.5|
|    109|  4.333333333333333|
|    105|  4.166666666666667|
|    110|             4.25|
|    106|             3.5|
|    104|  3.666666666666665|
+-----+

Execution Time AFTER Caching: 0.261327 seconds
```

Measure execution time after caching.

```
26/09/28 17:02:41 INFO DAGScheduler: Asked to cleanup jobs for query execution 4
+-----+
|movieId|    average_rating|
+-----+
|    108|             4.0|
|    101|             4.4|
|    103|             4.5|
|    107|             4.0|
|    102|             3.5|
|    109|  4.333333333333333|
|    105|  4.166666666666667|
|    110|             4.25|
|    106|             3.5|
|    104|  3.666666666666665|
+-----+

Execution Time AFTER Caching: 0.261327 seconds
```

Repartition the dataset.

```

Applying Repartitioning...
26/09/28 17:02:41 INFO DAGScheduler: Registering RDD 50 ($anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768) as input to shuffle 3
26/09/28 17:02:41 INFO DAGScheduler: Got map stage job 10 ($anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768) with 1 output partitions
26/09/28 17:02:41 INFO DAGScheduler: Final stage: ShuffleMapStage 13 ($anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768)
26/09/28 17:02:41 INFO DAGScheduler: Parents of final stage: List()
26/09/28 17:02:41 INFO DAGScheduler: Missing parents: List()
26/09/28 17:02:41 INFO DAGScheduler: Missing parents found for ShuffleMapStage 13: List()
26/09/28 17:02:41 INFO DAGScheduler: Submitting ShuffleMapStage 13 (MapPartitionsRDD[50] at $anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768), which has no missing parents
26/09/28 17:02:41 INFO MemoryStore: Block broadcast_16 stored as values in memory (estimated size 22.3 KiB, free 433.4 MiB)
26/09/28 17:02:41 INFO MemoryStore: Block broadcast_16_piece0 stored as bytes in memory (estimated size 10.0 KiB, free 433.4 MiB)
26/09/28 17:02:41 INFO SparkContext: Created broadcast 16 from broadcast at DAGScheduler.scala:1760
26/09/28 17:02:41 INFO DAGScheduler: Submitting 1 missing tasks from ShuffleMapStage 13 (MapPartitionsRDD[50] at $anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768) (first 15 tasks are for partitions Vector())
26/09/28 17:02:41 INFO TaskSchedulerImpl: Adding task set 13.0 with 1 tasks resource profile 0
26/09/28 17:02:41 INFO TaskSetManager: Starting task 0.0 in stage 13.0 (TID 10) (10.255.255.254,executor driver, partition 0, PROCESS_LOCAL, 10233 bytes)
26/09/28 17:02:41 INFO Executor: Running task 0.0 in stage 13.0 (TID 10)
26/09/28 17:02:41 INFO BlockManager: Found block rdd_29_0 locally
26/09/28 17:02:41 INFO CodeGenerator: Code generated in 8.739732 ms
26/09/28 17:02:41 INFO Executor: Finished task 0.0 in stage 13.0 (TID 10). 3019 bytes result sent to driver
26/09/28 17:02:41 INFO TaskSetManager: Finished task 0.0 in stage 13.0 (TID 10) in 41 ms on 10.255.255.254 (executor driver) (1/1)
26/09/28 17:02:41 INFO TaskSchedulerImpl: Removed TaskSet 13.0 whose tasks have all completed, from pool
26/09/28 17:02:41 INFO DAGScheduler: ShuffleMapStage 13 ($anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768) finished in 51 ms
26/09/28 17:02:41 INFO DAGScheduler: running: HashSet()
26/09/28 17:02:41 INFO DAGScheduler: waiting: HashSet()
26/09/28 17:02:41 INFO DAGScheduler: failed: HashSet()
Updated Number of Partitions: 4

```

Display updated partition count.

```

+-----+-----+
|movieId| average_rating|
+-----+-----+
| 108| 4.0|
| 101| 4.4|
| 103| 4.5|
| 105| 4.166666666666667|
| 106| 3.5|
| 109| 4.333333333333333|
| 104| 3.6666666666666665|
| 107| 4.0|
| 102| 3.5|
| 110| 4.25|
+-----+-----+

```

Execution Time AFTER Partitioning: 0.390049 seconds

Compare performance before and after optimization.

PERFORMANCE COMPARISON

```

=====
Initial Partitions      : 1
After Repartitioning    : 4

Execution Times:
Before Optimization     : 1.613389 seconds
After Caching           : 0.363679 seconds
After Partitioning      : 0.433090 seconds

Performance Improvement:
Caching Improvement     : 77.46%
Partitioning Improvement: 73.16%

```

Generate performance report.

```

=====
PERFORMANCE REPORT
=====

MovieFlix Technologies - Spark Performance Report

Dataset           : ratings.csv
Optimization Techniques: Caching and Partitioning

Initial Partitions : 1
Final Partitions   : 4

Execution Time Before Optimization : 1.613389 seconds
Execution Time After Caching       : 0.363679 seconds
Execution Time After Partitioning  : 0.433090 seconds

Caching Improvement      : 77.46%
Partitioning Improvement : 73.16%

Conclusion:
Caching stores frequently accessed DataFrame data in memory,
reducing the need to recompute the dataset.

Repartitioning distributes data across multiple partitions,
which can improve parallel processing for suitable workloads.

26/09/28 13:32:53 INFO MapPartitionsRDD: Removing RDD 29 from persistence list
26/09/28 13:32:53 INFO BlockManager: Removing RDD 29
26/09/28 13:32:53 INFO SparkContext: SparkContext is stopping with exitCode 0 from stop at NativeMethodAccessorImpl.java:0.
26/09/28 13:32:53 INFO SparkUI: Stopped Spark web UI at http://10.255.255.254:4040
26/09/28 13:32:53 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
26/09/28 13:32:54 INFO MemoryStore: MemoryStore cleared
26/09/28 13:32:54 INFO BlockManager: BlockManager stopped
26/09/28 13:32:54 INFO BlockManagerMaster: BlockManagerMaster stopped
26/09/28 13:32:54 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator stopped!
26/09/28 13:32:54 INFO SparkContext: Successfully stopped SparkContext (Uptime: 14166 ms)

Spark Session Stopped Successfully!
=====

```

Key Questions / Analysis

Q1. What is caching in Apache Spark and why does it improve performance?

Answer:

Caching is a performance optimization technique in Apache Spark that stores a DataFrame or RDD in memory after it is computed.

When the same data is used multiple times, Spark can reuse the cached data instead of reading and processing it again.

Example:

```
ratings_df.cache()
```

```
ratings_df.count()
```

The count() action triggers the caching.

Why does it improve performance?

Without caching:

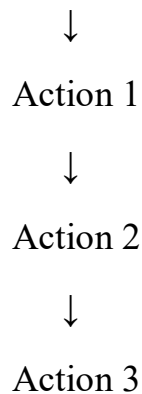
Data → Read → Process → Action 1

Data → Read → Process → Action 2

Data → Read → Process → Action 3

With caching:

Data → Read → Process → Cache



Therefore, caching is particularly useful when the same DataFrame is used for **multiple operations**.

Q2. What is the difference between `cache()` and `persist()`?

`cache()`

`cache()` stores the data using Spark's default storage level.

`df.cache()`

It is essentially a convenient shortcut for using the default persistence level.

`persist()`

`persist()` allows us to explicitly specify the storage level.

from pyspark import StorageLevel

`df.persist(StorageLevel.DISK_ONLY)`

Here the data is stored on disk rather than only in memory.

Difference

Example

`df.cache()`

versus:

`df.persist(StorageLevel.DISK_ONLY)`

Q3. How does Spark partition data internally?

Spark divides a large dataset into smaller pieces called **partitions**.

Each partition can be processed independently by Spark tasks.

Spark can process these partitions in parallel.

You can check the number of partitions using:

```
df.rdd.getNumPartitions()
```

For example:

Number of partitions: 4

Q4. Why does repartition() improve scalability?

repartition() redistributes data across a specified number of partitions.

Example:

```
df = df.repartition(4)
```

This creates four partitions.

This allows Spark to distribute work across multiple tasks and, in a cluster, across multiple executors.

Therefore, partitioning helps Spark **scale processing as the dataset grows**.

Note: repartition() itself causes a shuffle, so for very small datasets it can actually make execution slower. Its benefits are more relevant for larger workloads.

Q5. How does lazy evaluation affect Spark performance?

Spark uses **lazy evaluation**.

This means transformations are not executed immediately.

For example:

```
df2 = df.filter(df.rating > 4)
```

Spark does not immediately execute the filtering operation.

It builds a logical execution plan.

An action such as:

```
df2.show()
```

or:

```
df2.count()
```

triggers execution.

Benefit

Spark can optimize the execution plan before actually processing the data.

Transformations



Logical Plan



Optimization



Execution

This helps reduce unnecessary computation and improves performance.

Supplementary Problems:

Compare `cache()` and `persist(StorageLevel.DISK_ONLY)`. Repartition dataset into 2, 4 and 8 partitions and compare performance.

```

26/09/28 17:33:54 INFO DAGScheduler: ASKED to cleanup jobs for query execution 2
=====
|userId|movieId|rating|timestamp|
|-----|-----|-----|-----|
|1|101|4.0|964982703|
|1|102|3.5|964981247|
|1|103|5.0|964982224|
|2|101|4.5|964982931|
|2|104|3.0|964983815|
=====
only showing top 5 rows

=====
1. CACHE VS PERSIST(DISK_ONLY)
=====
26/09/28 17:33:54 INFO FileSourceStrategy: Pushed Filters:
26/09/28 17:33:54 INFO FileSourceStrategy: Post-Scan Filters: Set()
26/09/28 17:33:54 INFO MemoryStore: Block broadcast_6 stored as values in memory (estimated size 214.6 KiB, free 433.9 MiB)
26/09/28 17:33:54 INFO MemoryStore: Block broadcast_6_piece0 stored as bytes in memory (estimated size 39.6 KiB, free 433.9 MiB)
26/09/28 17:33:54 INFO SparkContext: Created broadcast 6 from count at NativeMethodAccessorImpl.java:0
26/09/28 17:33:54 INFO FileSourceScanExec: Planning scan with bin packing, max size: 4194304 bytes, open cost is considered as scanning 4194304 bytes.
26/09/28 17:33:54 INFO DAGScheduler: Got job 3 (count at NativeMethodAccessorImpl.java:0) with 1 output partitions
26/09/28 17:33:54 INFO DAGScheduler: Final stage: ResultStage 3 (count at NativeMethodAccessorImpl.java:0)
26/09/28 17:33:54 INFO DAGScheduler: Parents of final stage: List()
26/09/28 17:33:54 INFO DAGScheduler: Missing parents: List()
26/09/28 17:33:54 INFO DAGScheduler: Missing parents found for ResultStage 3: List()
26/09/28 17:33:54 INFO DAGScheduler: Submitting ResultStage 3 (FileScan csv [userId#17,movieId#18,rating#19,timestamp#20] Batched: false, DataFilters: [], Format: CSV, Location: InMemoryFileIndex(1 paths)[file:/home/gitanshu/spark_practical7/ratings.csv], PartitionFilters: [], PushedFilters: [], ReadSchema: struct<userId:int,movieId:int,rating:double,timestamp:int> MapPartitionsRDD[17] at count at NativeMethodAccessorImpl.java:0), which has no missing parents
26/09/28 17:33:54 INFO MemoryStore: Block broadcast_7 stored as values in memory (estimated size 16.2 KiB, free 434.1 MiB)
26/09/28 17:33:54 INFO MemoryStore: Block broadcast_7_piece0 stored as bytes in memory (estimated size 7.9 KiB, free 434.1 MiB)
26/09/28 17:33:54 INFO SparkContext: Created broadcast 7 from broadcast at DAGScheduler.scala:1760
26/09/28 17:33:54 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 3 (FileScan csv [userId#17,movieId#18,rating#19,timestamp#20] Batched: false, DataFilters: [], Format: CSV, Location: InMemoryFileIndex(1 paths)[file:/home/gitanshu/spark_practical7/ratings.csv], PartitionFilters: [], PushedFilters: [], ReadSchema: struct<userId:int,movieId:int,rating:double,timestamp:int> MapPartitionsRDD[17] at count at NativeMethodAccessorImpl.java:0) (first 15 tasks are for partitions Vector(0))
26/09/28 17:33:54 INFO TaskSchedulerImpl: Adding task set 3.0 with 1 tasks resource profile 0
26/09/28 17:33:54 INFO TaskSetManager: Starting task 0.0 in stage 3.0 (TID 3) (10.255.255.254,executor driver, partition 0, PROCESS_LOCAL, 10244 bytes)

```

```

Partitions: 2
Execution Time: 0.670117 seconds
26/09/28 17:33:56 INFO FileSourceStrategy: Pushed Filters:
26/09/28 17:33:56 INFO FileSourceStrategy: Post-Scan Filters: Set()
26/09/28 17:33:56 INFO MemoryStore: Block broadcast_21 stored as values in memory (estimated size 214.6 KiB, free 433.5 MiB)
26/09/28 17:33:56 INFO MemoryStore: Block broadcast_21_piece0 stored as bytes in memory (estimated size 39.6 KiB, free 433.5 MiB)
26/09/28 17:33:56 INFO SparkContext: Created broadcast 21 from $anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768
26/09/28 17:33:56 INFO FileSourceScanExec: Planning scan with bin packing, max size: 4194304 bytes, open cost is considered as scanning 4194304 bytes.
26/09/28 17:33:56 INFO DAGScheduler: Registering RDD 62 ($anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768) as input to shuffle 6
26/09/28 17:33:56 INFO DAGScheduler: Got map stage job 14 ($anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768) with 1 output partitions
26/09/28 17:33:56 INFO DAGScheduler: Final stage: ShuffleMapStage 22 ($anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768)
26/09/28 17:33:56 INFO DAGScheduler: Parents of final stage: List()
26/09/28 17:33:56 INFO DAGScheduler: Missing parents: List()
26/09/28 17:33:56 INFO DAGScheduler: Missing parents found for ShuffleMapStage 22: List()
26/09/28 17:33:56 INFO DAGScheduler: Submitting ShuffleMapStage 22 (MapPartitionsRDD[62] at $anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768), which has no missing parents
26/09/28 17:33:56 INFO MemoryStore: Block broadcast_22 stored as values in memory (estimated size 16.9 KiB, free 433.4 MiB)
26/09/28 17:33:56 INFO MemoryStore: Block broadcast_22_piece0 stored as bytes in memory (estimated size 8.2 KiB, free 433.4 MiB)
26/09/28 17:33:56 INFO SparkContext: Created broadcast 22 from broadcast at DAGScheduler.scala:1760
26/09/28 17:33:56 INFO DAGScheduler: Submitting 1 missing tasks from ShuffleMapStage 22 (MapPartitionsRDD[62] at $anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768) (first 15 tasks are for partitions Vector(0))
26/09/28 17:33:56 INFO TaskSchedulerImpl: Adding task set 22.0 with 1 tasks resource profile 0
26/09/28 17:33:56 INFO TaskSetManager: Starting task 0.0 in stage 22.0 (TID 15) (10.255.255.254,executor driver, partition 0, PROCESS_LOCAL, 10233 bytes)
26/09/28 17:33:56 INFO Executor: Running task 0.0 in stage 22.0 (TID 15)
26/09/28 17:33:56 INFO FileScanRDD: Reading File path: file:///home/gitanshu/spark_practical7/ratings.csv, range: 0-635, partition values: [empty row]
26/09/28 17:33:56 INFO Executor: Finished task 0.0 in stage 22.0 (TID 15). 1823 bytes result sent to driver

```

```

Partitions: 4
Execution Time: 0.339271 seconds
26/09/28 17:33:57 INFO FileSourceStrategy: Pushed Filters:
26/09/28 17:33:57 INFO FileSourceStrategy: Post-Scan Filters: Set()
26/09/28 17:33:57 INFO MemoryStore: Block broadcast_28 stored as values in memory (estimated size 214.6 KiB, free 433.3 MiB)
26/09/28 17:33:57 INFO MemoryStore: Block broadcast_28_piece0 stored as bytes in memory (estimated size 39.6 KiB, free 433.2 MiB)
26/09/28 17:33:57 INFO SparkContext: Created broadcast 28 from $anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768
26/09/28 17:33:57 INFO FileSourceScanExec: Planning scan with bin packing, max size: 4194304 bytes, open cost is considered as scanning 4194304 bytes.
26/09/28 17:33:57 INFO DAGScheduler: Registering RDD 83 ($anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768) as input to shuffle 10
26/09/28 17:33:57 INFO DAGScheduler: Got map stage job 19 ($anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768) with 1 output partitions
26/09/28 17:33:57 INFO DAGScheduler: Final stage: ShuffleMapStage 33 ($anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768)
26/09/28 17:33:57 INFO DAGScheduler: Parents of final stage: List()
26/09/28 17:33:57 INFO DAGScheduler: Missing parents: List()
26/09/28 17:33:57 INFO DAGScheduler: Missing parents found for ShuffleMapStage 33: List()
26/09/28 17:33:57 INFO DAGScheduler: Submitting ShuffleMapStage 33 (MapPartitionsRDD[83] at $anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768), which has no missing parents
26/09/28 17:33:57 INFO MemoryStore: Block broadcast_29 stored as values in memory (estimated size 17.0 KiB, free 433.2 MiB)
26/09/28 17:33:57 INFO MemoryStore: Block broadcast_29_piece0 stored as bytes in memory (estimated size 8.2 KiB, free 433.2 MiB)
26/09/28 17:33:57 INFO SparkContext: Created broadcast 29 from broadcast at DAGScheduler.scala:1760
26/09/28 17:33:57 INFO DAGScheduler: Submitting 1 missing tasks from ShuffleMapStage 33 (MapPartitionsRDD[83] at $anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768) (first 15 tasks are for partitions Vector(0))
26/09/28 17:33:57 INFO TaskSchedulerImpl: Adding task set 33.0 with 1 tasks resource profile 0
26/09/28 17:33:57 INFO TaskSetManager: Starting task 0.0 in stage 33.0 (TID 23) (10.255.255.254,executor driver, partition 0, PROCESS_LOCAL, 10233 bytes)
26/09/28 17:33:57 INFO Executor: Running task 0.0 in stage 33.0 (TID 23)
26/09/28 17:33:57 INFO FileScanRDD: Reading File path: file:///home/gitanshu/spark_practical7/ratings.csv, range: 0-635, partition values: [empty row]
26/09/28 17:33:57 INFO Executor: Finished task 0.0 in stage 33.0 (TID 23). 1870 bytes result sent to driver
26/09/28 17:33:57 INFO TaskSetManager: Finished task 0.0 in stage 33.0 (TID 23) in 67 ms on 10.255.255.254 (executor driver) (1/1)
26/09/28 17:33:57 INFO TaskSchedulerImpl: Removed TaskSet 33.0 whose tasks have all completed, from pool
26/09/28 17:33:57 INFO DAGScheduler: ShuffleMapStage 33 ($anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768) finished in 79 ms
26/09/28 17:33:57 INFO DAGScheduler: looking for newly runnable stages
26/09/28 17:33:57 INFO DAGScheduler: running: HashSet()
26/09/28 17:33:57 INFO DAGScheduler: waiting: HashSet()
26/09/28 17:33:57 INFO DAGScheduler: failed: HashSet()
26/09/28 17:33:57 INFO FileSourceStrategy: Pushed Filters:
26/09/28 17:33:57 INFO FileSourceStrategy: Post-Scan Filters: Set()
26/09/28 17:33:57 INFO MemoryStore: Block broadcast_30 stored as values in memory (estimated size 214.6 KiB, free 433.4 MiB)
26/09/28 17:33:57 INFO MemoryStore: Block broadcast_30_piece0 stored as bytes in memory (estimated size 39.6 KiB, free 433.4 MiB)
26/09/28 17:33:57 INFO SparkContext: Created broadcast 30 from $anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768
26/09/28 17:33:57 INFO FileSourceScanExec: Planning scan with bin packing, max size: 4194304 bytes, open cost is considered as scanning 4194304 bytes.
26/09/28 17:33:57 INFO DAGScheduler: Registering RDD 91 ($anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768) as input to shuffle 11

```

```

26/09/28 17:33:57 INFO DAGScheduler: waiting: HashSet()
26/09/28 17:33:57 INFO DAGScheduler: failed: HashSet()
26/09/28 17:33:57 INFO SparkContext: Starting job: $anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768
26/09/28 17:33:57 INFO DAGScheduler: Got job 23 ($anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768) with 1 output partitions
26/09/28 17:33:57 INFO DAGScheduler: Final stage: ResultStage 43 ($anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768)
26/09/28 17:33:57 INFO DAGScheduler: Parents of final stage: List(ShuffleMapStage 42)
26/09/28 17:33:57 INFO DAGScheduler: Missing parents: List()
26/09/28 17:33:57 INFO DAGScheduler: Missing parents found for ResultStage 43: List()
26/09/28 17:33:57 INFO DAGScheduler: Submitting ResultStage 43 (MapPartitionsRDD[100] at $anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768),
  which has no missing parents
26/09/28 17:33:57 INFO MemoryStore: Block broadcast_34 stored as values in memory (estimated size 14.1 KiB, free 433.2 MiB)
26/09/28 17:33:57 INFO MemoryStore: Block broadcast_34_piece0 stored as bytes in memory (estimated size 6.8 KiB, free 433.2 MiB)
26/09/28 17:33:57 INFO SparkContext: Created broadcast 34 from broadcast at DAGScheduler.scala:1760
26/09/28 17:33:57 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 43 (MapPartitionsRDD[100] at $anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768) (first 15 tasks are for partitions Vector(0))
26/09/28 17:33:57 INFO TaskSchedulerImpl: Adding task set 43.0 with 1 tasks resource profile 0
26/09/28 17:33:57 INFO TaskSetManager: Starting task 0.0 in stage 43.0 (TID 34) (10.255.255.254,executor driver, partition 0, NODE_LOCAL, 9631 bytes)
26/09/28 17:33:57 INFO Executor: Running task 0.0 in stage 43.0 (TID 34)
26/09/28 17:33:57 INFO ShuffleBlockFetcherIterator: Getting 1 (60.0 B) non-empty blocks including 1 (60.0 B) local and 0 (0.0 B) host-local and 0 (0.0 B) pu
sh-merged-local and 0 (0.0 B) remote blocks
26/09/28 17:33:57 INFO ShuffleBlockFetcherIterator: Started 0 remote fetches in 3 ms
26/09/28 17:33:57 INFO Executor: Finished task 0.0 in stage 43.0 (TID 34). 3895 bytes result sent to driver
26/09/28 17:33:57 INFO TaskSetManager: Finished task 0.0 in stage 43.0 (TID 34) in 13 ms on 10.255.255.254 (executor driver) (1/1)
26/09/28 17:33:57 INFO TaskSchedulerImpl: Removed TaskSet 43.0 whose tasks have all completed, from pool
26/09/28 17:33:57 INFO DAGScheduler: ResultStage 43 ($anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768) finished in 34 ms
26/09/28 17:33:57 INFO DAGScheduler: Job 23 is finished. Cancelling potential speculative or zombie tasks for this job
26/09/28 17:33:57 INFO TaskSchedulerImpl: Cancelling stage 43
26/09/28 17:33:57 INFO TaskSchedulerImpl: Killing all running tasks in stage 43: Stage finished
26/09/28 17:33:57 INFO DAGScheduler: Job 23 finished: $anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768, took 41.850215 ms
26/09/28 17:33:57 INFO DAGScheduler: Asked to cleanup jobs for query execution 7

```

```

Partitions: 8
Execution Time: 0.394943 seconds

```

Measure execution time for multiple Spark actions.

```

Action 1 - count()
Records: 38
Time: 0.109630 seconds
26/09/28 17:33:58 INFO CodeGenerator: Code generated in 7.89876 ms
26/09/28 17:33:58 INFO SparkContext: Starting job: showString at NativeMethodAccessorImpl.java:0
26/09/28 17:33:58 INFO DAGScheduler: Got job 27 (showString at NativeMethodAccessorImpl.java:0) with 1 output partitions
26/09/28 17:33:58 INFO DAGScheduler: Final stage: ResultStage 48 (showString at NativeMethodAccessorImpl.java:0)
26/09/28 17:33:58 INFO DAGScheduler: Parents of final stage: List()
26/09/28 17:33:58 INFO DAGScheduler: Missing parents: List()
26/09/28 17:33:58 INFO DAGScheduler: Missing parents found for ResultStage 48: List()
26/09/28 17:33:58 INFO DAGScheduler: Submitting ResultStage 48 (MapPartitionsRDD[117] at showString at NativeMethodAccessorImpl.java:0), which has no missin
g parents
26/09/28 17:33:58 INFO MemoryStore: Block broadcast_39 stored as values in memory (estimated size 24.0 KiB, free 433.0 MiB)
26/09/28 17:33:58 INFO MemoryStore: Block broadcast_39_piece0 stored as bytes in memory (estimated size 10.5 KiB, free 433.0 MiB)
26/09/28 17:33:58 INFO SparkContext: Created broadcast 39 from broadcast at DAGScheduler.scala:1760
26/09/28 17:33:58 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 48 (MapPartitionsRDD[117] at showString at NativeMethodAccessorImpl.java:0)
  (first 15 tasks are for partitions Vector(0))
26/09/28 17:33:58 INFO TaskSchedulerImpl: Adding task set 48.0 with 1 tasks resource profile 0
26/09/28 17:33:58 INFO TaskSetManager: Starting task 0.0 in stage 48.0 (TID 38) (10.255.255.254,executor driver, partition 0, PROCESS_LOCAL, 10244 bytes)
26/09/28 17:33:58 INFO Executor: Running task 0.0 in stage 48.0 (TID 38)
26/09/28 17:33:58 INFO BlockManager: Found block rdd_104_0 locally
26/09/28 17:33:58 INFO CodeGenerator: Code generated in 15.446528 ms
26/09/28 17:33:58 INFO CodeGenerator: Code generated in 9.071793 ms
26/09/28 17:33:58 INFO Executor: 1 block locks were not released by task 0.0 in stage 48.0 (TID 38)
  [rdd_104_0]
26/09/28 17:33:58 INFO Executor: Finished task 0.0 in stage 48.0 (TID 38). 2941 bytes result sent to driver
26/09/28 17:33:58 INFO TaskSetManager: Finished task 0.0 in stage 48.0 (TID 38) in 42 ms on 10.255.255.254 (executor driver) (1/1)
26/09/28 17:33:58 INFO TaskSchedulerImpl: Removed TaskSet 48.0 whose tasks have all completed, from pool
26/09/28 17:33:58 INFO DAGScheduler: ResultStage 48 (showString at NativeMethodAccessorImpl.java:0) finished in 48 ms
26/09/28 17:33:58 INFO DAGScheduler: Job 27 is finished. Cancelling potential speculative or zombie tasks for this job
26/09/28 17:33:58 INFO TaskSchedulerImpl: Cancelling stage 48
26/09/28 17:33:58 INFO TaskSchedulerImpl: Killing all running tasks in stage 48: Stage finished
26/09/28 17:33:58 INFO DAGScheduler: Job 27 finished: showString at NativeMethodAccessorImpl.java:0, took 51.207724 ms
26/09/28 17:33:58 INFO DAGScheduler: Asked to cleanup jobs for query execution 9

```

userId	movieId	rating	timestamp
1	101	4.0	964982703
1	102	3.5	964981247
1	103	5.0	964982224
2	101	4.5	964982931

```

Action 2 - show()
Time: 0.111860 seconds
26/09/28 17:33:58 INFO DAGScheduler: Registering RDD 122 ($anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768) as input to shuffle 15
26/09/28 17:33:58 INFO DAGScheduler: Got map stage job 28 ($anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768) with 1 output partitions
26/09/28 17:33:58 INFO DAGScheduler: Final stage: ShuffleMapStage 49 ($anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768)
26/09/28 17:33:58 INFO DAGScheduler: Parents of final stage: List()
26/09/28 17:33:58 INFO DAGScheduler: Missing parents: List()
26/09/28 17:33:58 INFO DAGScheduler: Missing parents found for ShuffleMapStage 49: List()
26/09/28 17:33:58 INFO DAGScheduler: Submitting ShuffleMapStage 49 (MapPartitionsRDD[122] at $anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:17
68), which has no missing parents
26/09/28 17:33:58 INFO MemoryStore: Block broadcast_40 stored as values in memory (estimated size 41.1 KiB, free 433.0 MiB)
26/09/28 17:33:58 INFO MemoryStore: Block broadcast_40_piece0 stored as bytes in memory (estimated size 16.0 KiB, free 433.0 MiB)
26/09/28 17:33:58 INFO SparkContext: Created broadcast 40 from broadcast at DAGScheduler.scala:1760
26/09/28 17:33:58 INFO DAGScheduler: Submitting 1 missing tasks from ShuffleMapStage 49 (MapPartitionsRDD[122] at $anonfun$withThreadLocalCaptured$1 at Comp
letableFuture.java:1768) (first 15 tasks are for partitions Vector(0))
26/09/28 17:33:58 INFO TaskSchedulerImpl: Adding task set 49.0 with 1 tasks resource profile 0
26/09/28 17:33:58 INFO TaskSetManager: Starting task 0.0 in stage 49.0 (TID 39) (10.255.255.254,executor driver, partition 0, PROCESS_LOCAL, 10233 bytes)
26/09/28 17:33:58 INFO Executor: Running task 0.0 in stage 49.0 (TID 39)
26/09/28 17:33:58 INFO BlockManager: Found block rdd_104_0 locally
26/09/28 17:33:58 INFO CodeGenerator: Code generated in 8.974984 ms
26/09/28 17:33:58 INFO Executor: Finished task 0.0 in stage 49.0 (TID 39). 3736 bytes result sent to driver
26/09/28 17:33:58 INFO TaskSetManager: Finished task 0.0 in stage 49.0 (TID 39) in 32 ms on 10.255.255.254 (executor driver) (1/1)
26/09/28 17:33:58 INFO TaskSchedulerImpl: Removed TaskSet 49.0 whose tasks have all completed, from pool
26/09/28 17:33:58 INFO DAGScheduler: ShuffleMapStage 49 ($anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768) finished in 40 ms
26/09/28 17:33:58 INFO DAGScheduler: looking for newly runnable stages

```

Use coalesce() and compare with repartition().

```

Using coalesce(2)
Partitions: 2
Records: 30
Time: 0.260824 seconds
26/09/28 17:33:58 INFO FileSourceStrategy: Pushed Filters:
26/09/28 17:33:58 INFO FileSourceStrategy: Post-Scan Filters: Set()
26/09/28 17:33:58 INFO MemoryStore: Block broadcast_51 stored as values in memory (estimated size 214.6 KiB, free 432.6 MiB)
26/09/28 17:33:58 INFO MemoryStore: Block broadcast_51_piece0 stored as bytes in memory (estimated size 39.6 KiB, free 432.6 MiB)
26/09/28 17:33:58 INFO SparkContext: Created broadcast 51 from $anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768
26/09/28 17:33:58 INFO FileSourceScanExec: Planning scan with bin packing, max size: 4194304 bytes, open cost is considered as scanning 4194304 bytes.
26/09/28 17:33:58 INFO DAGScheduler: Registering RDD 169 ($anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768) as input to shuffle 21
26/09/28 17:33:58 INFO DAGScheduler: Got map stage job 36 ($anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768) with 1 output partitions
26/09/28 17:33:58 INFO DAGScheduler: Final stage: ShuffleMapStage 63 ($anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768)
26/09/28 17:33:58 INFO DAGScheduler: Parents of final stage: List()
26/09/28 17:33:58 INFO DAGScheduler: Missing parents: List()
26/09/28 17:33:58 INFO DAGScheduler: Missing parents found for ShuffleMapStage 63: List()
26/09/28 17:33:58 INFO DAGScheduler: Submitting ShuffleMapStage 63 (MapPartitionsRDD[160] at $anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768), which has no missing parents
26/09/28 17:33:58 INFO MemoryStore: Block broadcast_52 stored as values in memory (estimated size 16.8 KiB, free 432.6 MiB)
26/09/28 17:33:58 INFO MemoryStore: Block broadcast_52_piece0 stored as bytes in memory (estimated size 8.2 KiB, free 432.6 MiB)
26/09/28 17:33:58 INFO SparkContext: Created broadcast 52 from broadcast at DAGScheduler.scala:1760
26/09/28 17:33:58 INFO DAGScheduler: Submitting 1 missing tasks from ShuffleMapStage 63 (MapPartitionsRDD[160] at $anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768) (first 15 tasks are for partitions Vector(0))
26/09/28 17:33:58 INFO TaskSchedulerImpl: Adding task set 63.0 with 1 tasks resource profile 0
26/09/28 17:33:58 INFO TaskSetManager: Starting task 0.0 in stage 63.0 (TID 48) (10.255.255.254,executor driver, partition 0, PROCESS_LOCAL, 10233 bytes)
26/09/28 17:33:58 INFO Executor: Running task 0.0 in stage 63.0 (TID 48)
26/09/28 17:33:58 INFO FileScanRDD: Reading File path: file:///home/gitanshu/spark_practical7/ratings.csv, range: 0-635, partition values: [empty row]
26/09/28 17:33:58 INFO Executor: Finished task 0.0 in stage 63.0 (TID 48). 1821 bytes result sent to driver
26/09/28 17:33:58 INFO TaskSetManager: Finished task 0.0 in stage 63.0 (TID 48) in 22 ms on 10.255.255.254 (executor driver) (1/1)
26/09/28 17:33:58 INFO TaskSchedulerImpl: Removed TaskSet 63.0 whose tasks have all completed, from pool
26/09/28 17:33:58 INFO DAGScheduler: ShuffleMapStage 63 ($anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768) finished in 27 ms
26/09/28 17:33:58 INFO DAGScheduler: looking for newly runnable stages
26/09/28 17:33:58 INFO DAGScheduler: running: HashSet()
26/09/28 17:33:58 INFO DAGScheduler: waiting: HashSet()
26/09/28 17:33:58 INFO DAGScheduler: failed: HashSet()
26/09/28 17:33:58 INFO FileSourceStrategy: Pushed Filters:
26/09/28 17:33:58 INFO FileSourceStrategy: Post-Scan Filters: Set()
26/09/28 17:33:58 INFO MemoryStore: Block broadcast_53 stored as values in memory (estimated size 214.6 KiB, free 432.6 MiB)
26/09/28 17:33:58 INFO MemoryStore: Block broadcast_53_piece0 stored as bytes in memory (estimated size 39.6 KiB, free 432.3 MiB)
26/09/28 17:33:58 INFO SparkContext: Created broadcast 53 from $anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768

```

```

which has no missing parents
26/09/28 17:33:58 INFO MemoryStore: Block broadcast_56 stored as values in memory (estimated size 14.1 KiB, free 432.6 MiB)
26/09/28 17:33:58 INFO MemoryStore: Block broadcast_56_piece0 stored as bytes in memory (estimated size 6.8 KiB, free 432.6 MiB)
26/09/28 17:33:58 INFO SparkContext: Created broadcast 56 from broadcast at DAGScheduler.scala:1760
26/09/28 17:33:58 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 69 (MapPartitionsRDD[174] at $anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768) (first 15 tasks are for partitions Vector(0))
26/09/28 17:33:58 INFO TaskSchedulerImpl: Adding task set 69.0 with 1 tasks resource profile 0
26/09/28 17:33:58 INFO TaskSetManager: Starting task 0.0 in stage 69.0 (TID 52) (10.255.255.254,executor driver, partition 0, NODE_LOCAL, 9631 bytes)
26/09/28 17:33:58 INFO Executor: Running task 0.0 in stage 69.0 (TID 52)
26/09/28 17:33:58 INFO ShuffleBlockFetcherIterator: Getting 2 (120.0 B) non-empty blocks including 2 (120.0 B) local and 0 (0.0 B) host-local and 0 (0.0 B) push-merged-local and 0 (0.0 B) remote blocks
26/09/28 17:33:58 INFO ShuffleBlockFetcherIterator: Started 0 remote fetches in 0 ms
26/09/28 17:33:58 INFO Executor: Finished task 0.0 in stage 69.0 (TID 52). 3895 bytes result sent to driver
26/09/28 17:33:58 INFO TaskSetManager: Finished task 0.0 in stage 69.0 (TID 52) in 14 ms on 10.255.255.254 (executor driver) (1/1)
26/09/28 17:33:58 INFO TaskSchedulerImpl: Removed TaskSet 69.0 whose tasks have all completed, from pool
26/09/28 17:33:58 INFO DAGScheduler: ResultStage 69 ($anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768) finished in 20 ms
26/09/28 17:33:58 INFO DAGScheduler: Job 39 is finished. Cancelling potential speculative or zombie tasks for this job
26/09/28 17:33:58 INFO TaskSchedulerImpl: Canceling stage 69
26/09/28 17:33:58 INFO TaskSchedulerImpl: Killing all running tasks in stage 69: Stage finished
26/09/28 17:33:58 INFO DAGScheduler: Job 39 finished: $anonfun$withThreadLocalCaptured$1 at CompletableFuture.java:1768, took 24.251296 ms
26/09/28 17:33:58 INFO DAGScheduler: Asked to cleanup jobs for query execution 12

Using repartition(2)
Partitions: 2
Records: 30
Time: 0.237833 seconds

```

Create larger ratings dataset and observe scalability.

```
=====
5. LARGE DATASET SCALABILITY
=====
26/09/28 13:54:33 INFO FileSourceStrategy: Pushed Filters:
26/09/28 13:54:33 INFO FileSourceStrategy: Post-Scan Filters: Set()
26/09/28 13:54:33 INFO FileSourceStrategy: Pushed Filters:
26/09/28 13:54:33 INFO FileSourceStrategy: Post-Scan Filters: Set()
26/09/28 13:54:33 INFO FileSourceStrategy: Pushed Filters:
26/09/28 13:54:33 INFO FileSourceStrategy: Post-Scan Filters: Set()
26/09/28 13:54:33 INFO FileSourceStrategy: Pushed Filters:
26/09/28 13:54:33 INFO FileSourceStrategy: Post-Scan Filters: Set()
26/09/28 13:54:33 INFO FileSourceStrategy: Pushed Filters:
26/09/28 13:54:33 INFO FileSourceStrategy: Post-Scan Filters: Set()
26/09/28 13:54:33 INFO FileSourceStrategy: Pushed Filters:
26/09/28 13:54:33 INFO FileSourceStrategy: Post-Scan Filters: Set()
26/09/28 13:54:33 INFO FileSourceStrategy: Pushed Filters:
26/09/28 13:54:33 INFO FileSourceStrategy: Post-Scan Filters: Set()
26/09/28 13:54:33 INFO FileSourceStrategy: Pushed Filters:
26/09/28 13:54:33 INFO FileSourceStrategy: Post-Scan Filters: Set()
26/09/28 13:54:33 INFO FileSourceStrategy: Pushed Filters:
26/09/28 13:54:33 INFO FileSourceStrategy: Post-Scan Filters: Set()
26/09/28 13:54:33 INFO FileSourceStrategy: Pushed Filters:
26/09/28 13:54:33 INFO FileSourceStrategy: Post-Scan Filters: Set()
26/09/28 13:54:33 INFO FileSourceStrategy: Pushed Filters:
26/09/28 13:54:33 INFO FileSourceStrategy: Post-Scan Filters: Set()
26/09/28 13:54:33 INFO MemoryStore: Block broadcast_57 stored as values in m
emory (estimated size 214.6 KiB, free 432.4 MiB)
```

1. performance.

Tools and Technologies:

- **Programming Language:** Python 3
- **Processing Framework:** Apache Spark
- **Spark API:** PySpark
- **Data Processing:** Spark DataFrames
- **Input Data Format:** CSV
- **Data Source:** Movie Ratings Dataset
- **Data Analysis:** Spark SQL / DataFrame Operations
- **Aggregation:** `groupBy()`, `agg()`, `avg()`
- **Performance Measurement:** `time.perf_counter()`
- **Caching:** `cache()`, `persist()`
- **Storage Level:** `StorageLevel.DISK_ONLY`
- **Partitioning:** `repartition()`, `coalesce()`
- **Partition Monitoring:** `getNumPartitions()`

- **Operating System:** Ubuntu Linux / WSL2
- **Execution Environment:** Local Spark (local[*])
- **Package Management:** Python pip
- **Virtual Environment:** Python venv
- **Development Environment:** Ubuntu Terminal / Nano
- **Execution Tool:** spark-submit
- **Dataset File:** ratings.csv

Conclusion:

The practical demonstrated how caching and partitioning can be used to optimize Spark application performance. Caching allows frequently reused data to be stored and accessed without repeatedly recomputing it, while partitioning distributes data into smaller units that can be processed in parallel. The practical also compared `cache()` with `persist(StorageLevel.DISK_ONLY)` and `repartition()` with `coalesce()`. The execution-time experiments showed how Spark performance can vary depending on the number of partitions, storage method, dataset size, and workload. The practical also demonstrated that optimization benefits are more noticeable for larger datasets and repeated Spark operations.